

Practice Question Paper

Waves

Student Name:

Date: 10.03.2026

Total Marks: 35

Instructions

Answer all questions. Show all relevant working clearly. Where appropriate, define the physical quantities you use and write down the equations before substituting values. Give explanations in full sentences where required.

Question 1: Is the particle moving with the wave?

A wave travels along a rope from left to right. A small ribbon is tied to one point on the rope. As the wave passes, the ribbon moves up and down through a maximum distance of 3 cm from its equilibrium position. The wave has a frequency of 5 Hz.

- (a) State the amplitude of the wave.
- (b) Does the ribbon travel along the rope with the wave? Explain your answer.
- (c) Calculate the number of complete oscillations made by the ribbon in 4 s.

Hint: Use the definition of amplitude carefully. For part (c), remember that frequency means the number of oscillations per second.

Question 2: Reading wavelength from a pattern

In a snapshot of a transverse wave, the distance from one crest to the next crest is 1.8 m. The frequency of the wave is 2.5 Hz.

- (a) Determine the wavelength.
- (b) Find the distance from a crest to the next trough.
- (c) Calculate the wave speed.

Hint: Crest to crest is one full wavelength. Crest to trough is half a wavelength.

Question 3: From observation to frequency

A floating cork on water completes 12 oscillations in 9 s. The distance between two neighbouring crests is 0.75 m.

- (a) Calculate the frequency of the wave.
- (b) Determine the period of the wave.
- (c) Calculate the speed of the wave.

Hint: First find the frequency from total oscillations divided by total time. Then use

$$T = \frac{1}{f}$$

and finally

$$v = f\lambda.$$

Question 4: Deciding the wave type and using the mathematics

A sound wave travels through air. The air molecules vibrate back and forth parallel to the direction of travel. The wave has a frequency of 340 Hz and a wavelength of 1.0 m.

- (a) State whether this wave is transverse or longitudinal, and explain your reasoning.
- (b) Calculate the wave speed.
- (c) If the frequency doubled while the wave remained in the same medium and the wave speed stayed constant, describe what would happen to the wavelength.

Hint: For part (c), use

$$v = f\lambda.$$

If v stays constant and f increases, then λ must decrease.

Question 5: Two students disagree

A wave in a spring has wavelength 4 m and frequency 3 Hz.

Student A says: “A particle in the spring travels along the spring at 12 m s^{-1} .”

Student B says: “The wave travels along the spring at 12 m s^{-1} , but the particles only oscillate about their equilibrium positions.”

- (a) Calculate the wave speed.
- (b) State which student is correct.
- (c) Explain the difference between wave motion and particle motion.

Hint: Use

$$v = f\lambda$$

first. Then recall the key idea that waves transfer energy without transferring matter.

Question 6: Working backwards

A wave travels with speed 6 m s^{-1} and has frequency 4 Hz .

- (a) Calculate the wavelength.
- (b) Determine the time taken for one complete oscillation.
- (c) In one second, how many wavelengths pass a fixed point?

Hint: Use

$$v = f\lambda$$

to find the wavelength and

$$T = \frac{1}{f}$$

to find the period. For part (c), connect your reasoning to the meaning of frequency.

Question 7: Comparing two waves

Wave P and Wave Q travel through the same medium.

$$\text{Wave P: } f = 2 \text{ Hz}, \quad \lambda = 5 \text{ m}$$

$$\text{Wave Q: } f = 4 \text{ Hz}, \quad \lambda = 2.5 \text{ m}$$

- (a) Calculate the speed of each wave.
- (b) State what you notice about the two results.
- (c) Explain why this makes sense if both waves travel in the same medium.

Hint: Calculate both speeds before drawing a conclusion. Think about what usually determines wave speed.

End of Paper